# WebSocket / VCH Compatibility — Analysis & Proposed Fix

Why the new bot's chats do not appear in the API ticketing "Visitors Chat History" (VCH) page, and how to fix it safely

| | | |
|---|---|---|
| **Prepared for:** Engineering review / approval | **Scope:** new_bot vs kebo_fe + lumira | **Date:** 25 Jun 2026 |

## Summary (corrected after deeper trace)

The VCH page does **not** read over the WebSocket. It reads chat history from the **KEBO_ENGAGE_SESSIONS** database table. Our bot **already writes to that same table**. So the real incompatibility is **not the WebSocket transport** — it is a **mismatch in the row we write** versus what lumira (the canonical writer) writes and what the page expects. That makes the fix **additive, low-risk, and confined to the history-write path**; the live chat WebSocket does not need to change.

## How the VCH page actually gets its data

- Frontend calls *POST /dashboard-svc/insert/get-latest-conversations.htm* with *{ sessionId, botId }*.
- That reads **GHN_PROD_BAK.KEBO_ENGAGE_SESSIONS** — lumira's *fetch_last_n_messages()* shows the contract: filter by *session_id + conversation_id*, *sender IN (USER, BOT, AGENT, INTERNAL)*, order by *TIMESTAMP*.
- Rows are **written** by lumira's *insert_data_into_db()* after each turn. Our bot writes the equivalent row via *engage_session_service.write_turn()*, gated to kebo hotels by *is_kebo_hotel()*.

> ■ Conclusion: the WebSocket (STOMP vs raw) is a **red herring for history**. If our row lands in KEBO_ENGAGE_SESSIONS with the right keys, it shows in VCH — regardless of transport.

## Where our write diverges from lumira's canonical write

| | | | |
|---|---|---|---|
| **is_conversation_active / use_for_context** | not written | we write = 1 | If those columns are absent in the table the Java page reads, EVERY insert throws — and we swallow the error → zero rows → VCH empty. Top suspect. |
| **entity_id** | resolved server-side (FMS mapping) | taken from widget frame metadata → often NULL | Row does not attach to a property/entity → not ticketable / filtered out. |
| **widget_id** | = Bot_Id | from settings map | Must equal the botId the page queries with, or rows are invisible. |
| **conversation_id** | Java / lumira id | our rotating UUID | If the page threads by its own id, history will not group. |
| **TIMESTAMP** | DB default | DB default | Fine IF the column has DEFAULT CURRENT_TIMESTAMP — needs confirming. |

Red row = highest-probability root cause for "nothing shows in VCH".

## Proposed fix (additive — /ws/chat untouched)

1. **Match lumira's exact column set.** Remove *is_conversation_active* / *use_for_context* from our INSERT unless a *DESCRIBE KEBO_ENGAGE_SESSIONS* confirms they exist. Removes the "every insert silently fails" risk.

2. **Resolve entity_id server-side** (via *ticketing_property_entity_map_json* / the scoped property) instead of trusting widget metadata. Never write NULL entity_id for a kebo hotel.
3. **Verify widget_id == the dashboard's botId** and that group_id is the real Kepsla id (already via *resolve_widget_group*).
4. **Use the page's conversation_id** as the threading key — confirm whether the ticketing page generates its own id and adopt it if so.
5. **Confirm TIMESTAMP has a server default**; set it explicitly in the INSERT if not, so ordering and the "latest" window work.

> **Why nothing of ours breaks:** every change is inside the KES writer + entity resolution, already gated by *is_kebo_hotel()* → Nexoria and other tenants are untouched. The WebSocket protocol, chat flow, HTTP fallback, message dedup, and suggestion chips are all unchanged.

## Verification plan

- ■ Read-only *DESCRIBE GHN_PROD_BAK.KEBO_ENGAGE_SESSIONS* — confirm columns + defaults.
- ■ Send one test turn for a kebo hotel; confirm a row lands with correct entity_id / widget_id / conversation_id / sender / TIMESTAMP.
- ■ Open the ticketing VCH page for that session; confirm the conversation renders in order.

## Appendix — transport difference (for context only)

Their widget speaks **STOMP over WebSocket** (subscribe */topic/room/{sessionId}*, publish */app/kebo-chat.sendMessage*) to a Java broker; the answer engine is lumira over plain HTTP. Ours is a **raw FastAPI WebSocket** at */ws/chat* with JSON request/response. This difference only matters for **live agent-takeover presence**, not for VCH history, and is therefore out of scope for this fix. It can be scoped separately if real-time takeover on the ticketing page is required.